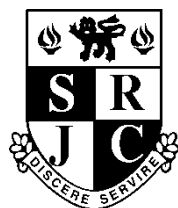


CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2014

BIOLOGY
Higher 2

Paper 3

9648

25 August 2013 / Monday
2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

FOR EXAMINER'S USE	
1	/13
2	/12
3	/15
4 (Planning)	/12
5 (Essay)	/20
TOTAL	/72

Answer **all** questions in all the sections.

At the end of examination,

1. Fasten all your work securely together;

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 12 printed pages

Question 1

The umbilical cord links a baby to the mother and carries nutrients and gases for the growth of baby. In the past, a huge portion of the baby cord was discarded. Now it is believed that the umbilical cord blood stem cells can possibly help cure several diseases. As such, many individuals now store their umbilical cord after giving birth. The cord is rich in two types of stem cells, they are hematopoietic stem cells and mesenchymal cells.

(a) Describe two features of stems cells that normal adult cells do not possess. [2]

.....

.....

.....

.....

(b) The two types of stem cells found in the umbilical cord are considered by scientists to be multipotent. Explain what this means and suggest how such stem cells could potentially be used to cure human diseases without involving gene therapy. [3]

.....

.....

.....

.....

.....

.....

Sometimes, DNA from stem cells is extracted for use in further medical research. The amounts extracted are usually very small and are seldom enough for practical use. A technique that can be employed to increase the amounts of extracted DNA is the Polymerase Chain Reaction (PCR). **Fig 1.1** on the following page illustrates what happens to DNA during this process with a specific focus on the temperatures involved.

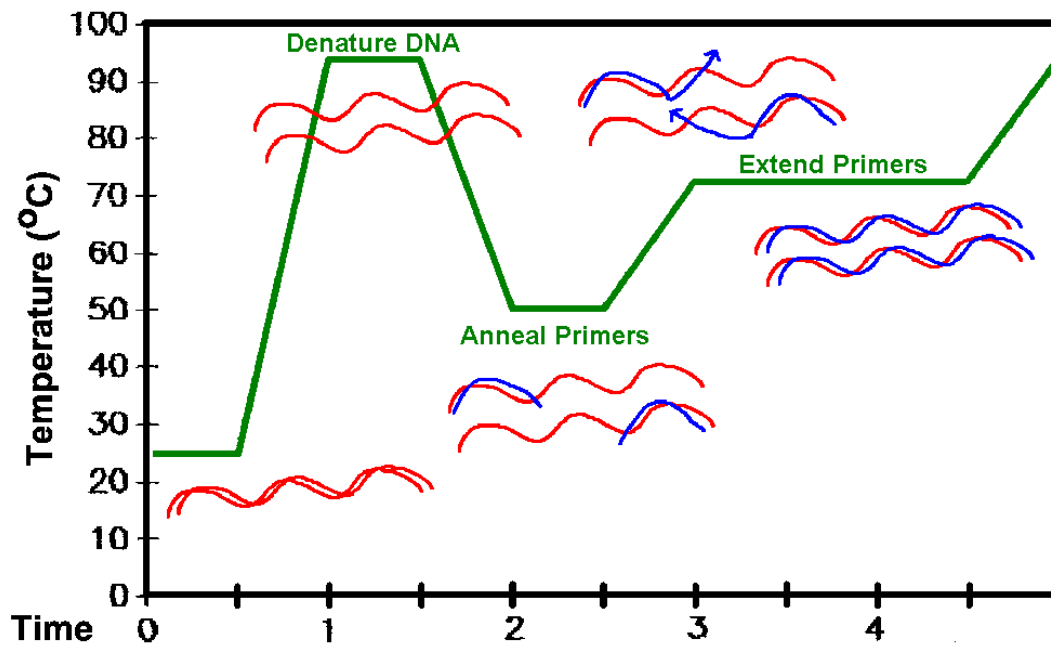


Fig 1.1

(c) Explain the significance of using different temperatures to this process. [3]

.....

.....

.....

.....

.....

.....

(d) Describe the key differences between this process and DNA replication in human cells. [3]

.....

.....

.....

.....

.....

.....

- (e) This process will not be able to proceed without the use of large amounts of specific primers. Explain why large amounts are needed and why they need to be specific. [2]

.....

.....

.....

.....

[Total: 13 marks]

Question 2

Huntington's disease (HD) is a lethal neurodegenerative disease that is inherited as an autosomal dominant condition. As the appearance of symptoms may occur later in life (as late as at age 40), patients with HD may unknowingly pass on the mutant HD allele to their children.

A radioactive G8 probe revealed two RFLP variants that are linked to the HD allele which can be used in the screening of HD. **Fig 2.1** shows the two RFLP variants upstream to the HD allele and the G8 probe. The arrows show the position of *Hind*III restriction sites. The numbers represent kilobases.

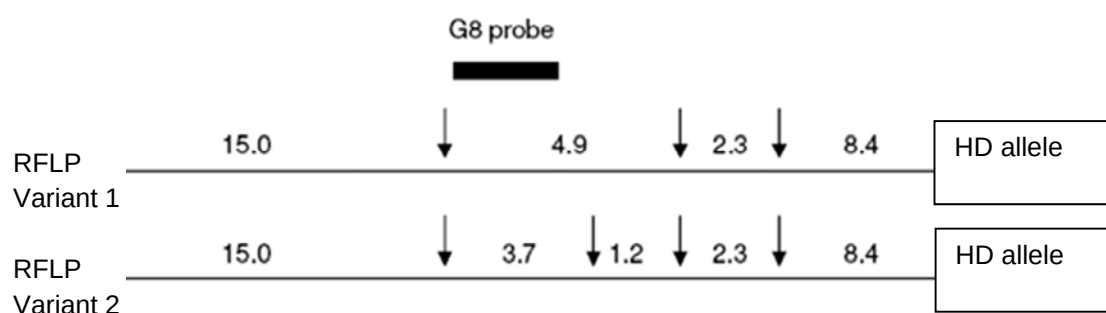


Fig 2.1

Fig 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.

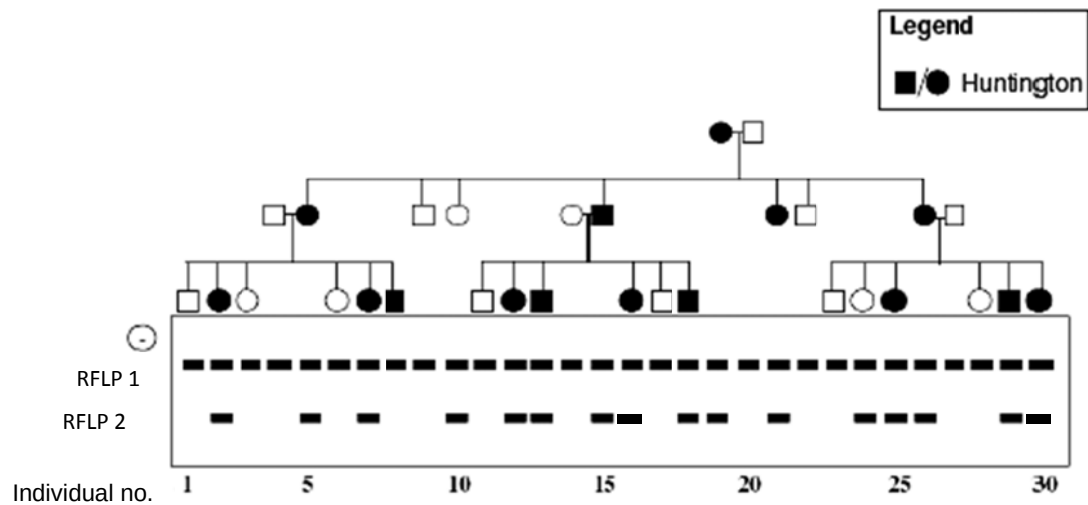


Fig 2.2

With reference to **Fig 2.1** and **2.2**,

(a) outline how the DNA band patterns of the 30 individuals were obtained. [4]

.....

.....

.....

.....

.....

.....

.....

.....

(b) identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

.....

.....

.....

.....

(c) suggest two possible reasons for the difference in the DNA band patterns among unaffected individuals. [4]

.....

.....

.....

.....

.....

.....

.....

.....

(d) discuss the effectiveness of using this RFLP marker in the screening of HD.[2]

.....

.....

.....

.....

[Total: 12 marks]

Question 3

The following **Fig 3.1** illustrates the prevalence of **genetically modified crops** in the world over a number of years.

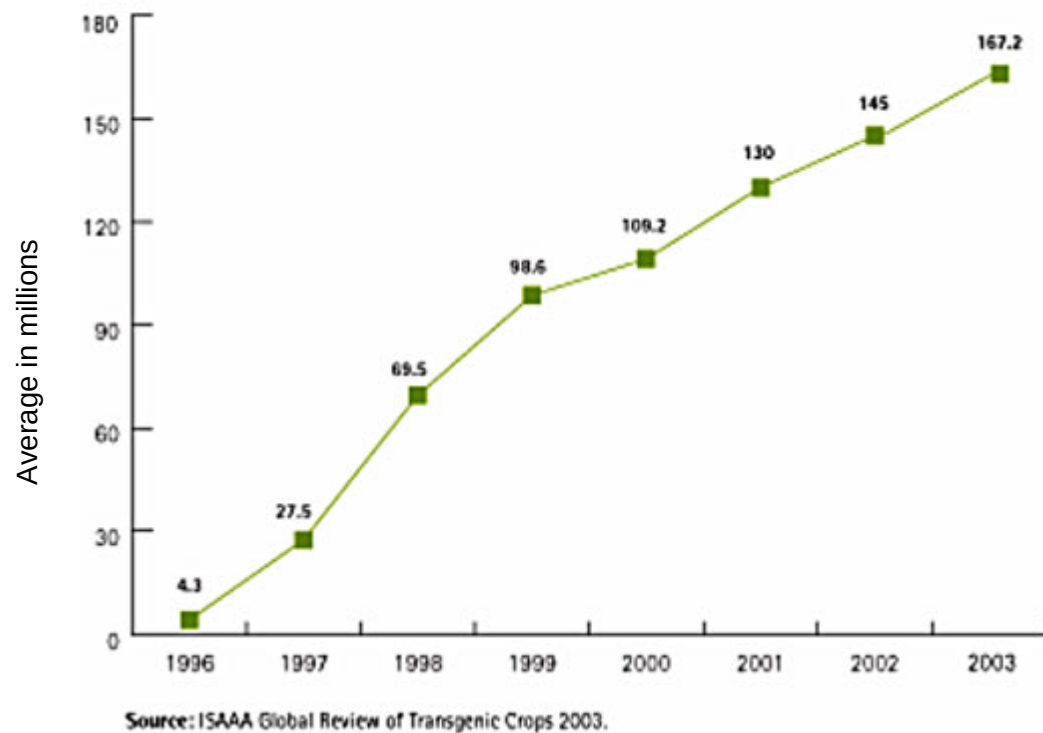


Fig 3.1

(a) Describe what you understand by the term '**genetically modified crops**'. [2]

.....

.....

.....

.....

(b) With reference to **Fig 3.1**, describe the growth in genetically modified crops between the years 1998 and 2003. [2]

.....

.....

.....

.....

(c) Explain, with the use of a named example, how genetically modified crops has led to positive outcomes for mankind. [4]

.....

.....

.....

.....

.....

.....

.....

.....

(d) In spite of possible positive outcomes, genetically modified crops have yet to be fully accepted my many countries. In fact, many individuals are openly against the use of genetic modification for food products. Comment. [2]

.....

.....

.....

.....

.....

.....

Fig 3.2 illustrates how plant tissue culture techniques may be leveraged upon to bring about genetic modification in plant tissues.

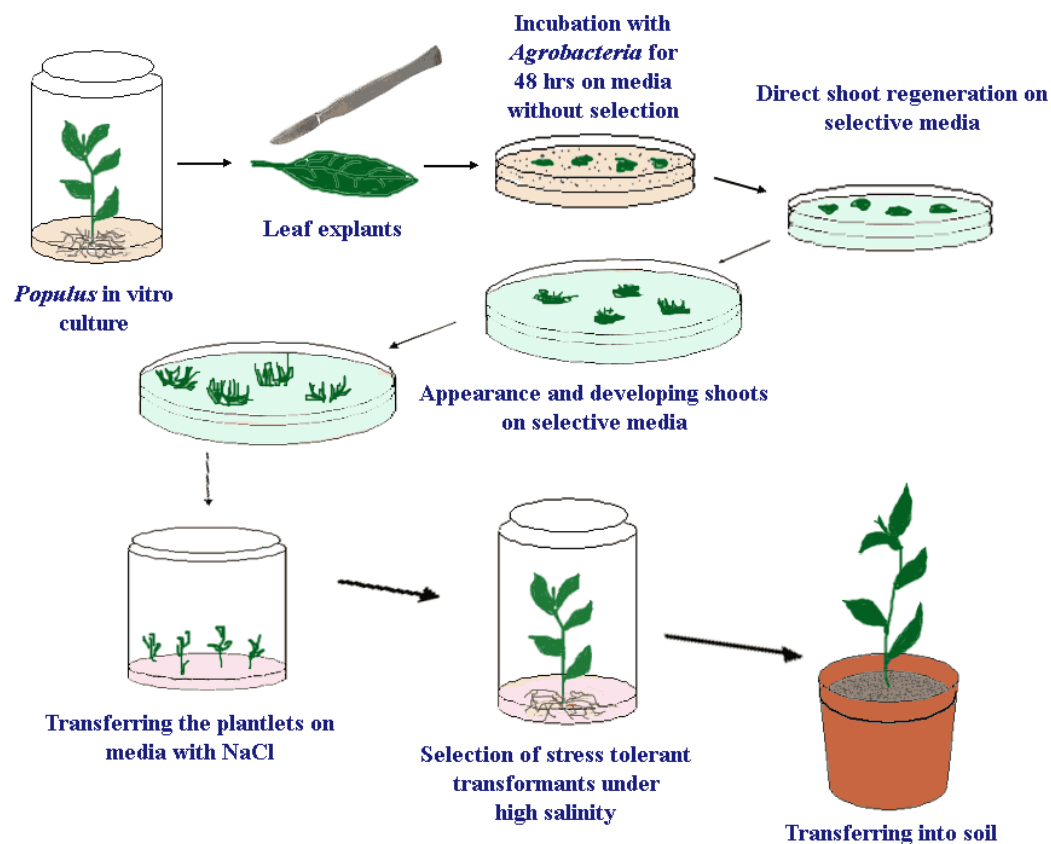


Fig 3.2

(e) Using the information from **Fig 3.2**, describe the process of cloning plants through plant tissue culture techniques. [3]

.....

.....

.....

.....

.....

.....

.....

- (f) *Agrobacterium* is commonly used in the above process as depicted in **Fig 3.2**.
Explain the role of *Agrobacterium* in this process. [2]

.....

.....

.....

.....

.....

.....

[Total: 15 marks]

Question 4 – Planning Question

In mammals, the products of digestion of food are absorbed from the intestinal cavity.

To investigate the uptake of glucose, a short piece of intestine is turned inside out, so that the absorptive surface is on the outside. The ends of the intestine are tied to form a bag containing a buffer solution, which is then placed in a boiling tube with a known concentration of glucose solution.

The concentration of glucose in various solutions can be determined by using a colour standard. This can be prepared by reacting a series of glucose solutions of different concentrations with Benedict's solution.

Using this information and your own knowledge, design an experiment to determine the effect of glucose concentration on the rate of uptake of glucose by the piece of intestine.

You must use:

- 10 % glucose solution
- Distilled water
- Piece of intestine tied to form a bag containing buffer solution
- Benedict's solution
- 5 glucose solutions at 0.1%, 0.5%, 1%, 2% and 4%

You should select from the following apparatus:

- Normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Timer e.g. stopwatch
- Syringes
- Bunsen burner with tripod, gauze and bench mat
- thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams if necessary,
- identify the dependent and independent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment

Question 5 - Free-response question

Write your answer on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

- (a) Describe the genetic basis of cystic fibrosis and compare viral and non-viral delivery systems in the treatment of cystic fibrosis. [8]
- (b) Discuss the social and ethical considerations for the use of gene therapy. [6]
- (c) Explain the problems associated with the expression of eukaryotic genes in prokaryotes and suggest how these problems can be overcome. [6]